

Lesson 3: SAS Visual Text Analytics Nodes



3.0 Lesson Overview



Lesson Overview

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



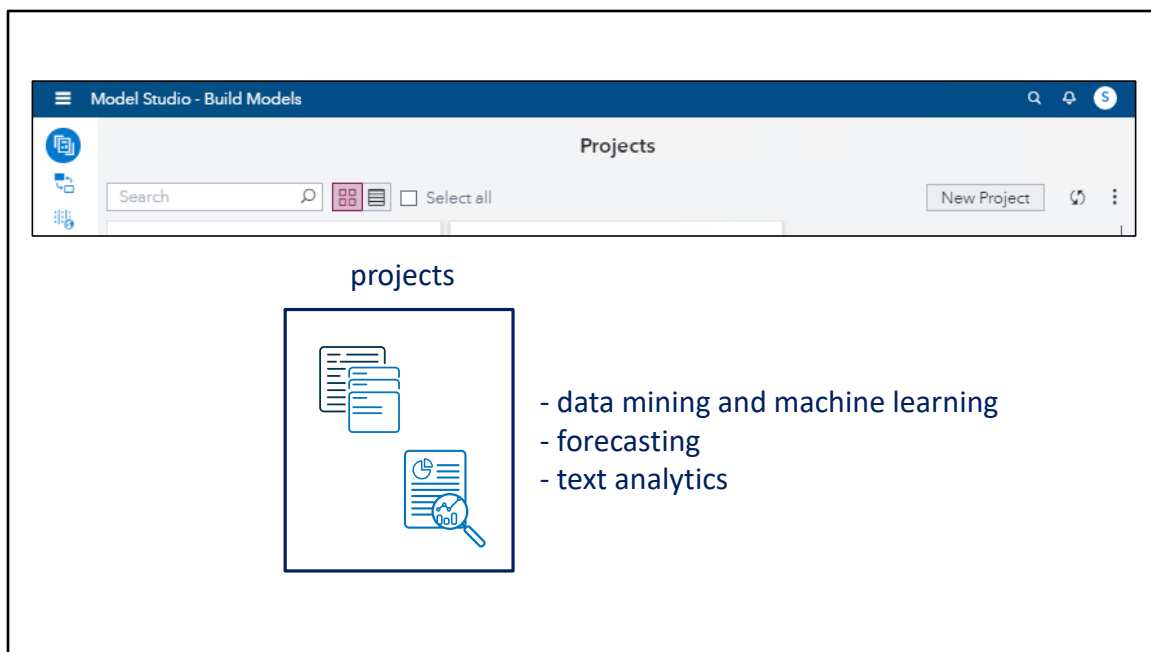
3.1 Projects



Creating Projects

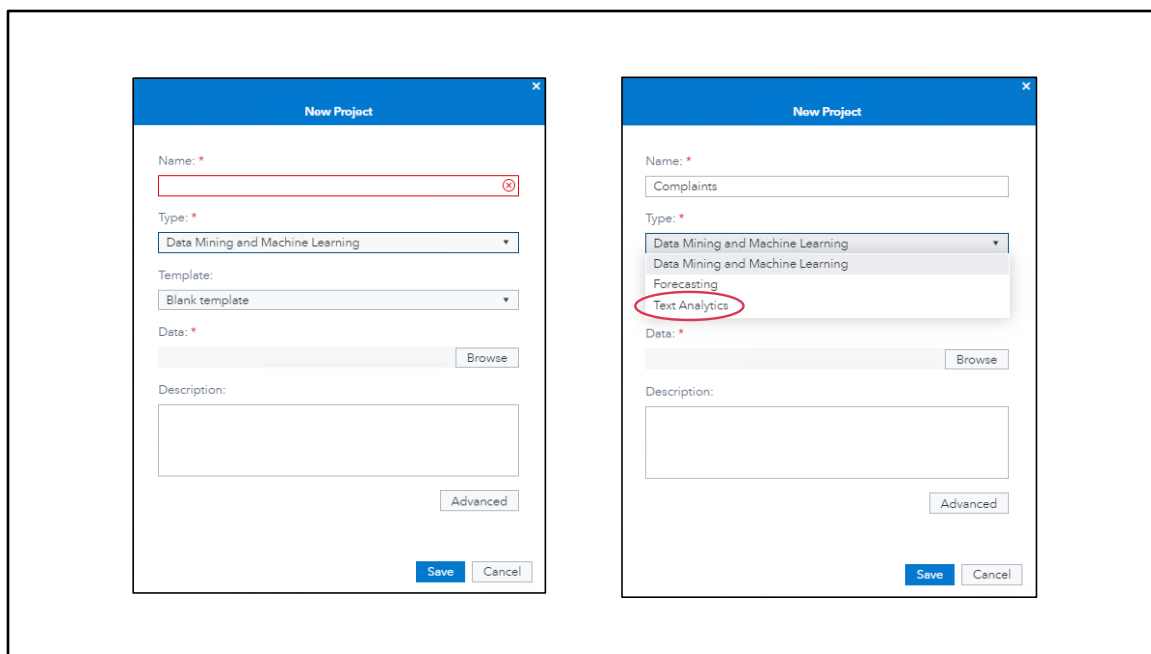


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You access SAS Visual Text Analytics through Model Studio. In Model Studio, you create *projects*, which are containers for your data and analysis. A project contains the input data, options, and analysis tasks. Three types of projects are supported in Model Studio and are related to three analysis areas: data mining and machine learning, forecasting, and text analytics. Model Studio is designed so that you can create and run multiple projects simultaneously.

To create a new project, click the **New Project** button in Model Studio.



SAS Visual Text Analytics supports the Text Analytics project type. When you select **Text Analytics**, you work with concepts, terms, sentiment, topics, and categories. Concepts, topics, and categories are built based on the terms in the project document collection.

The image shows a 'New Project' dialog box on the left and a vertical pipeline diagram on the right. The dialog box has fields for Name (filled with 'Complaints'), Type (set to 'Text Analytics'), Template (set to 'Text Analytics: Generate Concepts, Topics, a...'), Data (with a 'Browse' button), and Description. It includes 'Save' and 'Cancel' buttons at the bottom. The pipeline diagram on the right consists of six nodes connected by downward arrows: 'Data' (teal), 'Concepts' (orange), 'Text Parsing' (orange), 'Sentiment' (green), 'Topics' (green), and 'Categories' (blue). Each node has a small icon and a vertical ellipsis menu on its right side.

The default visual text analytics pipeline places the five nodes in sequence after a Data node. The default ordering is Concepts, Text Parsing, Sentiment, Topics, and Categories. If a Sentiment node appears before a Topics or Categories node, then sentiment information will be available in the document tables for those nodes.

You set the options for a node by selecting the node and changing the settings that are shown to the right of the pipeline. Each node is described in detail later.

You can also provide a project description.

DrugReport

Role:

Text

Language detection method:

Manual

Project Language:

English

Arabic

Chinese

Croatian

Czech

Danish

Dutch

English

Farsi

Finnish

French

German

Greek

Hebrew

Hindi

Hungarian

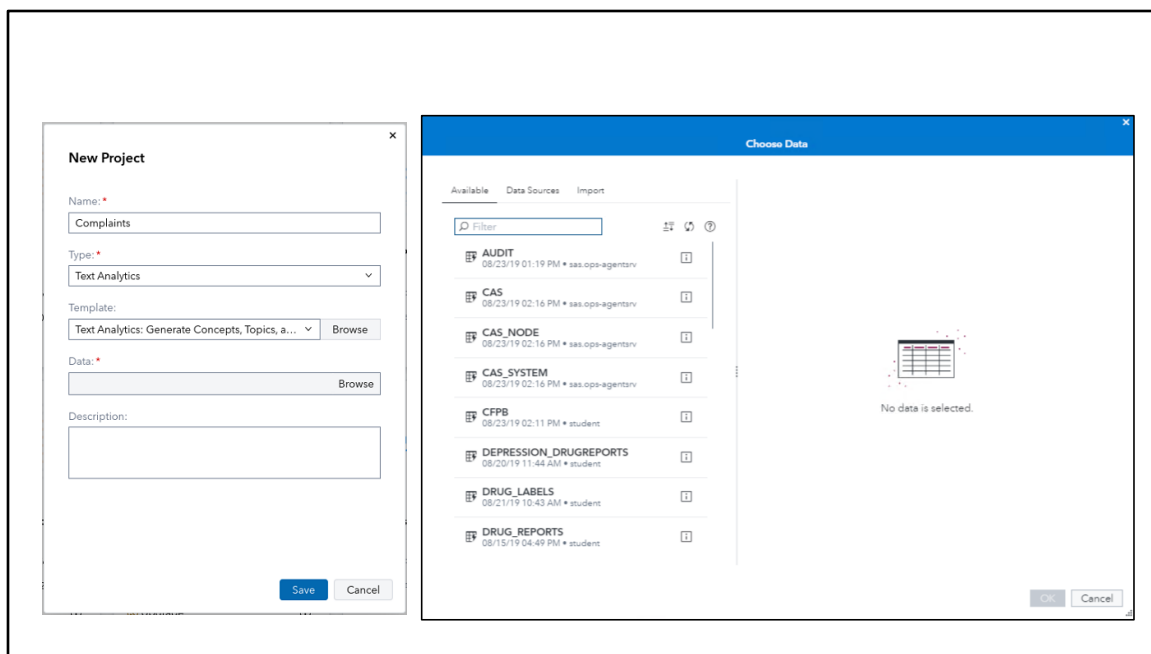
Indonesian

Italian

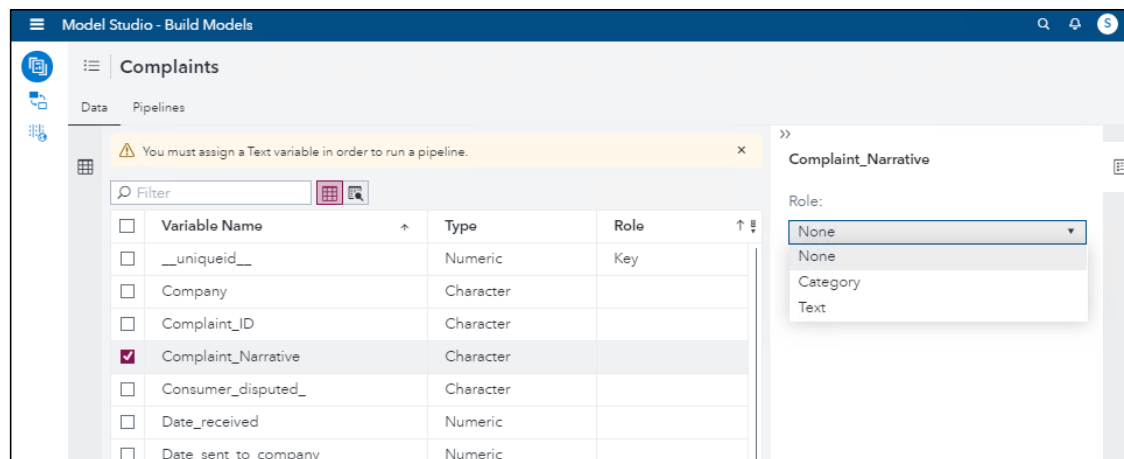
30+ Languages

Language for a Visual Text Analytics project is now set on the Data tab.

Over 30 languages are supported, and new languages are being added all the time. There are thousands of spoken languages, but many languages have a very small population of native speakers.

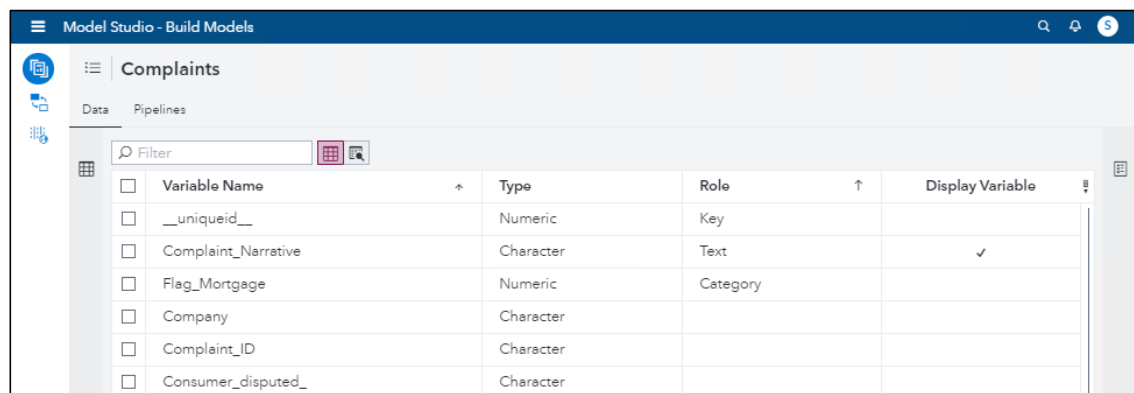


You can browse to find the analysis data set. The data set can already reside in memory, or you can import an existing SAS table to copy it into memory. You can also access a document collection and convert the documents into text format, then compile the documents into an analysis data set and copy it into memory.



The data set you select must have at least one variable with a role of Text. The variable with the role of Text contains the documents used for text analysis.

A categorical variable with the role Category is optional, but if one is identified, linguistic rules will be derived to classify each document into a level of the categorical variable.



☐	Variable Name	Type	Role	Display Variable
☐	__uniqueid__	Numeric	Key	
☐	Complaint_Narrative	Character	Text	✓
☐	Flag_Mortgage	Numeric	Category	
☐	Company	Character		
☐	Complaint_ID	Character		
☐	Consumer_disputed_	Character		

In this example, **Complaint_Narrative** contains the documents used for the analysis. The **Flag_Mortgage** variable has value *one* for mortgage complaints, and the value *zero* otherwise. Linguistic rules will be derived to classify a document as *zero* or *one* with respect to the mortgage flag.

☐	Variable Name	Type	Role	Display Variable
☐	__uniqueid__	Numeric	Key	
☐	Company	Character		
☐	Complaint_ID	Character		
☐	Complaint_Narrative	Character		
☐	Consumer_disputed_	Character		
☐	Date_received	Numeric		
☐	Date_sent_to_company	Numeric		
☐	DocLength	Numeric		
☐	Flag_Communication	Numeric		
☐	Flag_DataCollection	Numeric		

And, although a variable having a role of Key is required, one is supplied automatically by the software.

Resource

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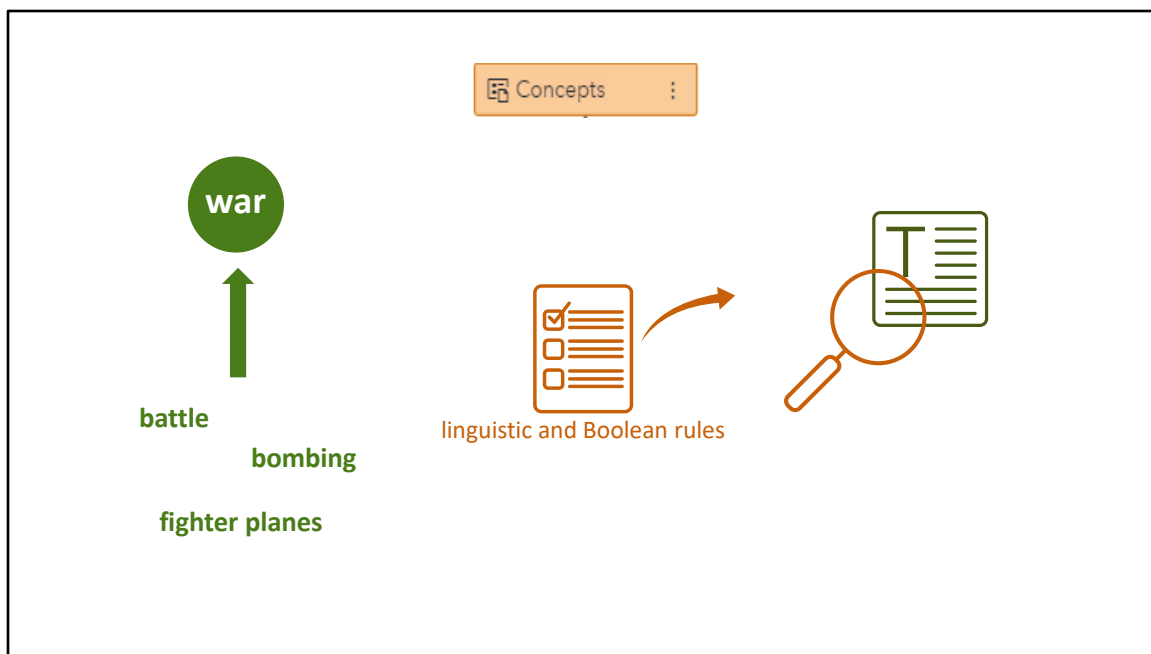
3.2 Concepts



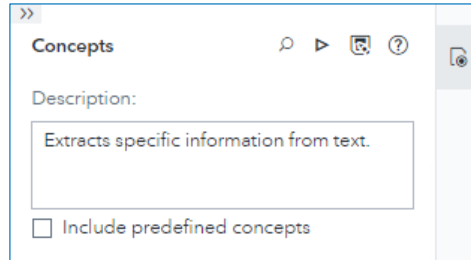
Exploring the Concepts Node



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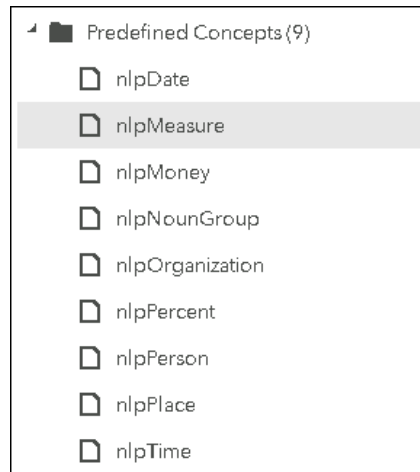
Concepts are useful for analyzing information in context. You can write rules for recognizing concepts that are important to you and thereby create custom concepts. For example, you can specify that the concept *war* is identified when the terms *battle*, *bombing*, and *fighter planes* are encountered in text.



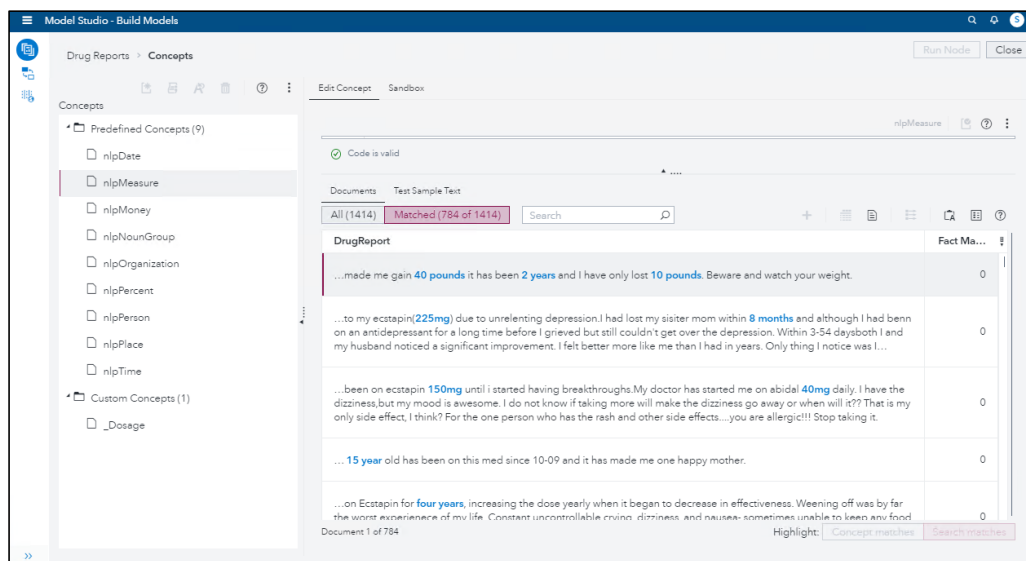
To examine the concepts settings, select the **Concepts** node.

The Concepts node has a single property: Include predefined concepts.

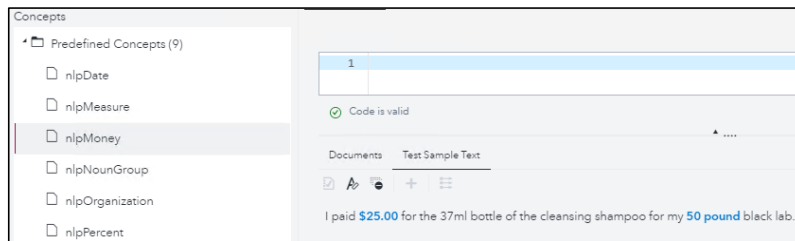
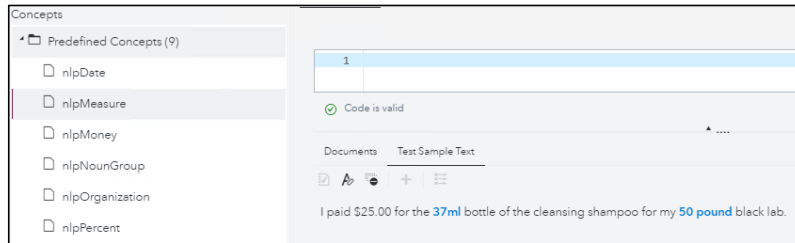
If you do not select the **Include predefined concepts** check box, and you do not provide custom concepts, then you can delete the Concepts node from the pipeline.



SAS Visual Text Analytics provides nine predefined concepts, which are concepts whose rules are already written. Predefined concepts save time by providing you with commonly used concepts and their definitions, such as *Measure* or *Organization*. You cannot edit predefined concepts or their rules, but you can append additional rules in the provided code editor. The predefined concepts supported by SAS Visual Text Analytics were created based on multiple language relevance.



When you click on a concept such as **nlpMeasure**, and select **Matched**, Model Studio presents the documents containing the concept, and highlights the terms that satisfy the concept. The concept rules that define a predefined concept are not shown. However, the definition of a custom concept can contain a reference to another concept, whether it's custom or predefined.



Although you can't see the rules that define a predefined concept, you can use the Test Sample Text feature to investigate what terms are highlighted in a given document related to the predefined concept rules.

3.3 Terms



Exploring the Text Parsing Node



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Start list

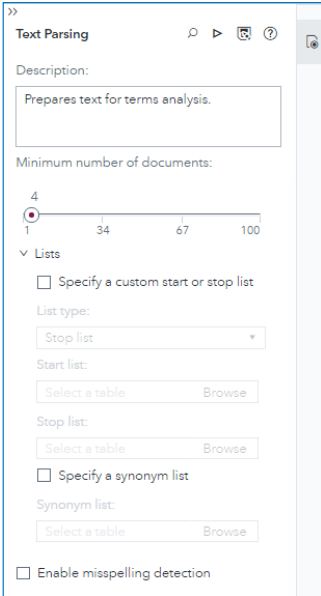
- terms to include
- inclusion dictionary

Stop list

- terms to exclude
- exclusion dictionary







Synonym list

- pairs of words to treat as single terms
- required format



The Text Parsing node has several properties. You use Start lists and Stop lists to control which terms are used and which are ignored.

A Start list is a data set that contains a list of terms that should be included in the analysis. Sentiment, topics, and categories use only the terms in the Start list. A Start list is also called an inclusion dictionary.

A Stop list is a data set that contains a list of terms that should be excluded from the analysis. You can use Stop lists to exclude terms that contain little information or that are extraneous to your Visual Text Analytics tasks. Sentiment, topics, and categories do not use the terms in the Stop list. A Stop list is also called an exclusion dictionary.

A default Stop list is provided for English (en_stoplist.sashdat) and for all supported languages.

Dictionary tables that are used as Start or Stop lists can be SAS tables. They do not need to be CAS tables.

The Minimum number of documents specifies how many documents a term must appear in to be allowed to be in a Start list. A term that is in a Start list, or is absent from a Stop list, will be placed in the Stop list if it appears in fewer than the minimum number of documents.

A synonym list is a SAS data set that identifies pairs of words that should be treated as single terms for the purpose of analysis. Synonym lists are stored in data sets and have a required format.

The Text Parsing node also has a spell-checking feature.

Text Parsing: Kept and Dropped Terms



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term

- label for group of characters, strings, or patterns that represent a single concept
- building blocks for topics, term maps, and category rules
- associated role is either blank, identifies term's part of speech, or identifies concept

token

- sequence of characters without separator

surface form

- variant of a term in a matched subset of text
- include inflected forms, synonyms, misspellings, and other ways to reference

stemming

- process of identifying inflected forms for nouns and verbs

Let's go over some vocabulary. A *term* is defined as a label for a group of characters, strings, or patterns that represent a single concept (an idea) as defined by underlying rules or algorithms. In SAS Visual Text Analytics, a term is the basic building block for topics, term maps, and category rules. Each term has an associated role that either is blank, identifies the term's part of speech, or identifies the concept that is associated with the term.

For alphabetic writing systems, a word is a *token* (a sequence of characters without a separator), and a term contains one or more words. There is a subtle difference between a term and a concept. When the Concept node appears before the Text Parsing node, a concept is eligible to be placed in the term table.

A *surface form* is a variant of a term that is in a matched subset of text. Surface forms can include inflected forms, synonyms, misspellings, and other ways of referring to a term.

Stemming is the process of identifying inflected forms for nouns and verbs. For example, *going* is an inflected form of *go*.

Model Studio - Build Models

Drug Reports > Text Parsing - Manage Terms

Kept Terms (1522)

Term *	Role	Documents	Frequency ↓
☐ not	ADV	658	1174
☐ > take	V	677	1105
☐ > depression	N	492	616
☐ > feel	V	371	518
☐ > year	N	395	502
☐ > drug	N	341	485
☐ > day	N	344	479
☐ > medication	N	304	446

Dropped Terms (6849)

Term *	Role	Documents	Frequency ↓
☐ i	PRO	1081	5003
☐ > be	V	1129	4183
☐ > have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386
☐ it	PRO	882	2100
☐ a	DET	792	1655

Start List

Stop List

To examine the terms that are used in the analysis, right-click the Text Parsing node and select Open.

When you open the Text Parsing node, you see the Kept Terms and Dropped Terms tables. Kept Terms are terms that reside in the Start list, and Dropped Terms reside in the Stop list. By default, both lists are sorted in descending order based on the frequency that each term appears.

You can move a kept term to the Dropped Terms pane by selecting the term and clicking the **Drop terms** icon, and you can move a dropped term to the Kept Terms pane by selecting the term and clicking the **Keep terms** icon.

If you make changes to the terms, you must rerun the project to see the effects of your changes.

Model Studio - Build Models

Drug Reports > Text Parsing - Manage Terms

Kept Terms (1522) Filter

Term	Role	Documents	Frequency
☐ not	ADV	658	1174
☐ take	V	677	1105
☒ depression	N	492	616
☐ feel	V	371	518
☐ year	N	395	502
☐ drug	N	341	485
☐ day	N	344	479
☐ medication	N	304	446

Dropped Terms (6849) Filter

Term	Role	Documents	Frequency
☐ i	PRO	1081	5003
☐ be	V	1129	4183
☐ have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386
☐ it	PRO	882	2100
☐ a	DET	792	1655

Documents

All (1414) Matched (492 of 1414) Search

DrugReport

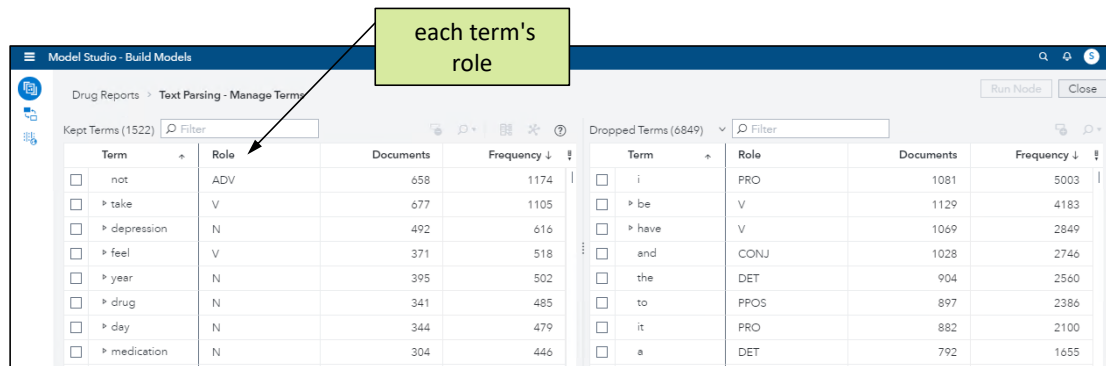
...due to unrelenting **depression**. I had lost my sister mom within 8 months and although I had been on an antidepressant for a long time before I grieved but still couldn't get over the **depression**. Within 3-5 days both I and my husband noticed a significant improvement. I felt better more like me than I had in years. Only thing I notice was I sometimes mix up words. Anybody else do that? Not bad enough for me to stop med tho.

...an abyss of **depression**, one which I have never experienced before, I had to go back on the prexifan and taper myself off for a month, even then I was having a continuous panic attack with heart pounding, no eating for 4-5 days, severe sense of loneliness, guilt, and poor self-worth. It was absolutely miserable that lasted for about 6-10 days. I have family support, but needed to see a psychiatrist (still waiting for visit). Be very cautious going on this med. It's great for a couple weeks, but very exhausting.

Document 1 of 492

Highlight: Term matches Search matches

You can view documents that contain a kept term by selecting the term and selecting **Matched** in the Documents pane.



Model Studio - Build Models

Drug Reports > Text Parsing - Manage Terms

Kept Terms (1522) | Filter

Term *	Role	Documents	Frequency ↓
☐ not	ADV	658	1174
☐ > take	V	677	1105
☐ > depression	N	492	616
☐ > feel	V	371	518
☐ > year	N	395	502
☐ > drug	N	341	485
☐ > day	N	344	479
☐ > medication	N	304	446

Dropped Terms (6849) | Filter

Term *	Role	Documents	Frequency ↓
☐ i	PRO	1081	5003
☐ > be	V	1129	4183
☐ > have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386
☐ it	PRO	882	2100
☐ a	DET	792	1655

The Role column displays each term's role, if one can be determined. A role could be a part of speech or a concept. To view the synonyms that were assigned to a term, click the triangle that appears next to that term.

Synonym list

- pairs of words to treat as single terms
- required format



TERM

TERMROLE

PARENT

PARENTROLE

Recall that a synonym list is a SAS data set that identifies pairs of words that should be treated as single terms for the purposes of analysis, and these lists have a required format.

You must include the following variables:

TERM, which contains a term to treat as a synonym of the parent, and PARENT, which contains the representative term to which the term should be assigned.

TERMROLE, which enables you to specify that the synonym is assigned only when the term occurs in the role that is specified in this variable. A term role is a function that is performed by a term in a particular context. Term roles include part-of-speech roles, entity roles, and user-defined roles.

And PARENTROLE, which enables you to specify the role of the parent.

Model Studio - Build Models

Drug Reports > Text Parsing - Manage Terms

Run Node Close

Kept Terms (1522) Filter

Term	Role	Documents	Frequency
☐ not	ADV	658	1174
☐ > take	V	677	1105
☐ > depression	N	492	616
☐ > feel	V	371	518
☐ > year	N	395	502
☐ > drug	N	341	485
☐ > day	N	344	479
☐ > medication	N	304	446

Dropped Terms (6849) Filter

Term	Role	Documents	Frequency
☐ i	PRO	1081	5003
☐ > be	V	1129	4183
☐ > have	V	1069	2849
☐ and	CONJ	1028	2746
☐ the	DET	904	2560
☐ to	PPOS	897	2386
☐ it	PRO	882	2100
☐ a	DET	792	1655



a

an

the

and

or

but

By default, words that provide little or no value are excluded from analysis. Examples of these words include the articles a, an, and the, and conjunctions such as and, or, and but. Other terms that are specific to your document collection, but provide little or no value, are also identified and excluded.

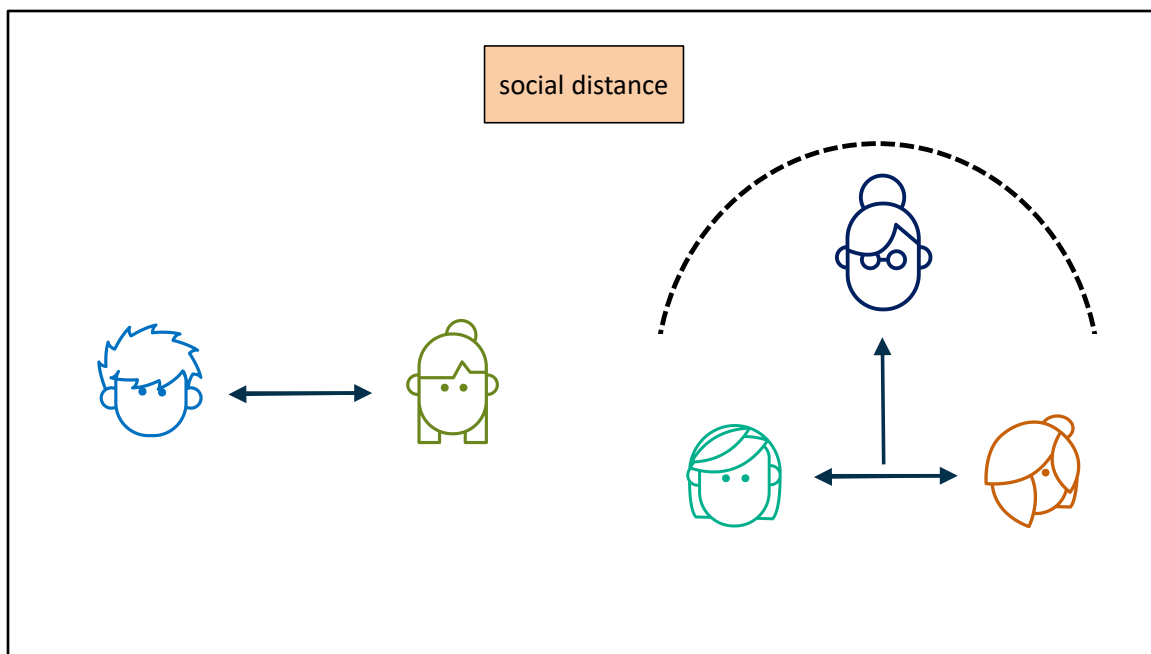
Similarity Scores and Terms Maps



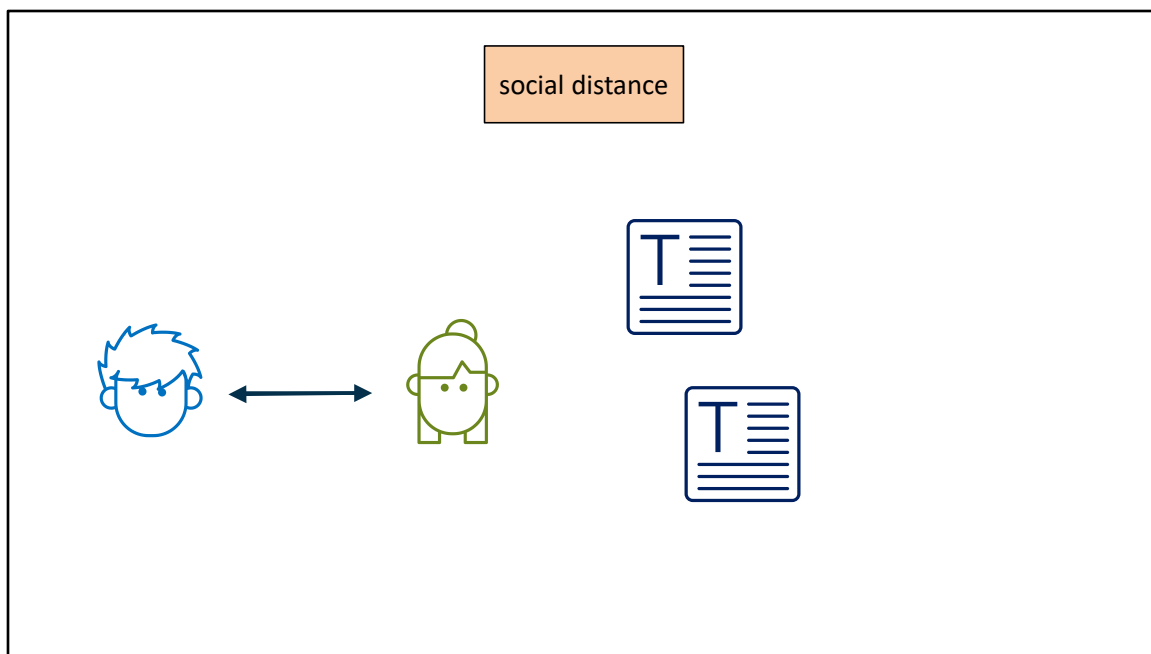
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	Term ↑	Similarity ↓	Role	Documents	Frequency
☒	▷ depression	1.000	N	492	616
☐	major	0.672	A	64	65
☐	anxiety	0.655	N	187	231
☐	major depression	0.640	nlpNounGroup	27	27
☐	immensely	0.614	ADV	4	4
☐	▷ satisfy	0.578	V	17	17
☐	chronic	0.576	A	22	23
☐	chronic pain	0.559	nlpNounGroup	5	6
☐	▷ combine	0.554	V	10	10
☐	depression	0.551	PN	9	10

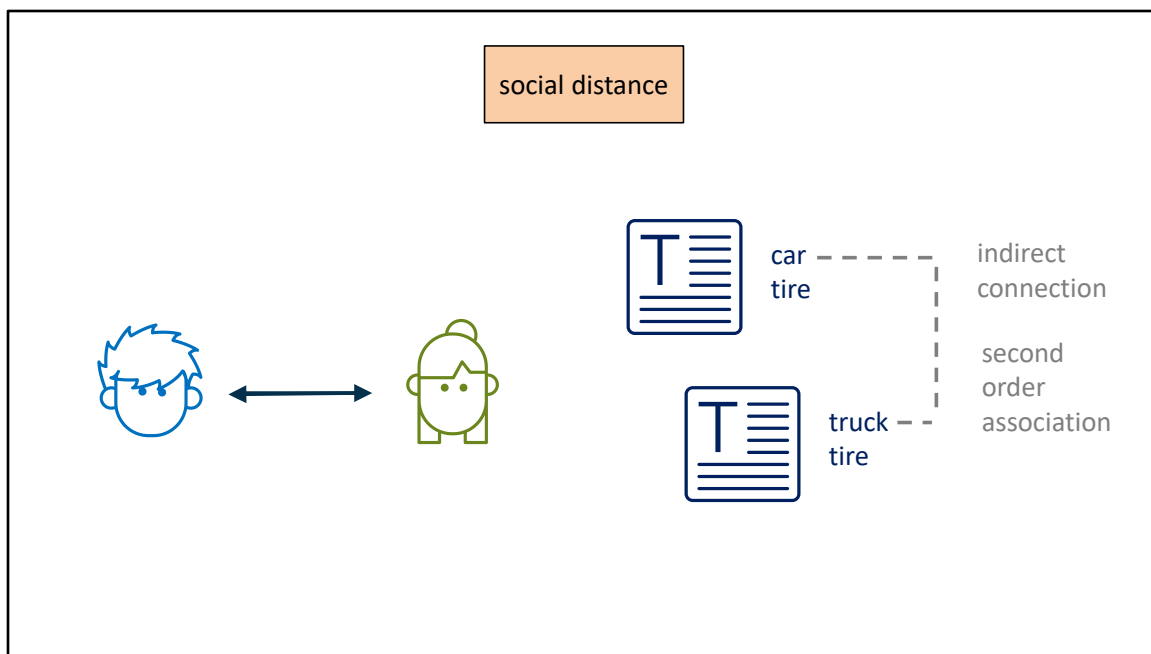
You can also view similarity scores for a term. Similarity scores are derived based on the linguistic distance between words as measured by co-occurrence frequencies. Social network metrics can be used to derive a social distance between two people based on community membership, and similar arguments can be used to derive the distance between words based on document membership (or co-occurrence). Thus, from a social network perspective, the personas *depression* and *major* have the most social interaction, while *anxiety*, *major depression*, and *immensely* are also in the social mix.



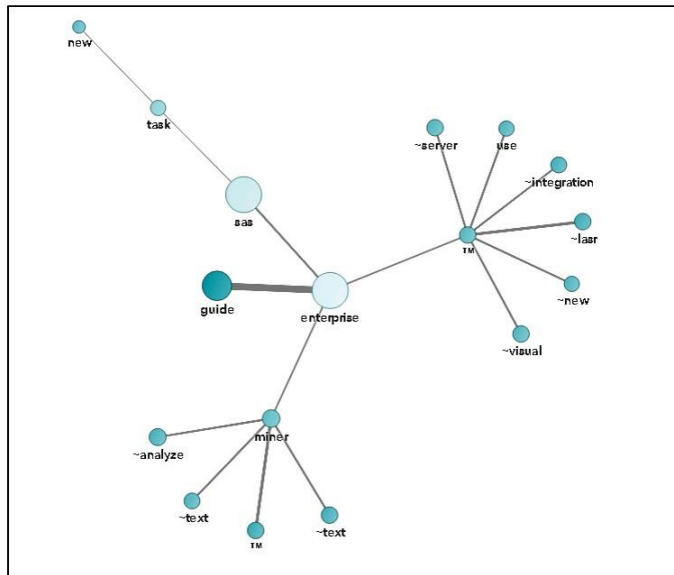
Social distance relates to two individuals interacting directly by speaking with each other or interacting indirectly by communicating through a third party.



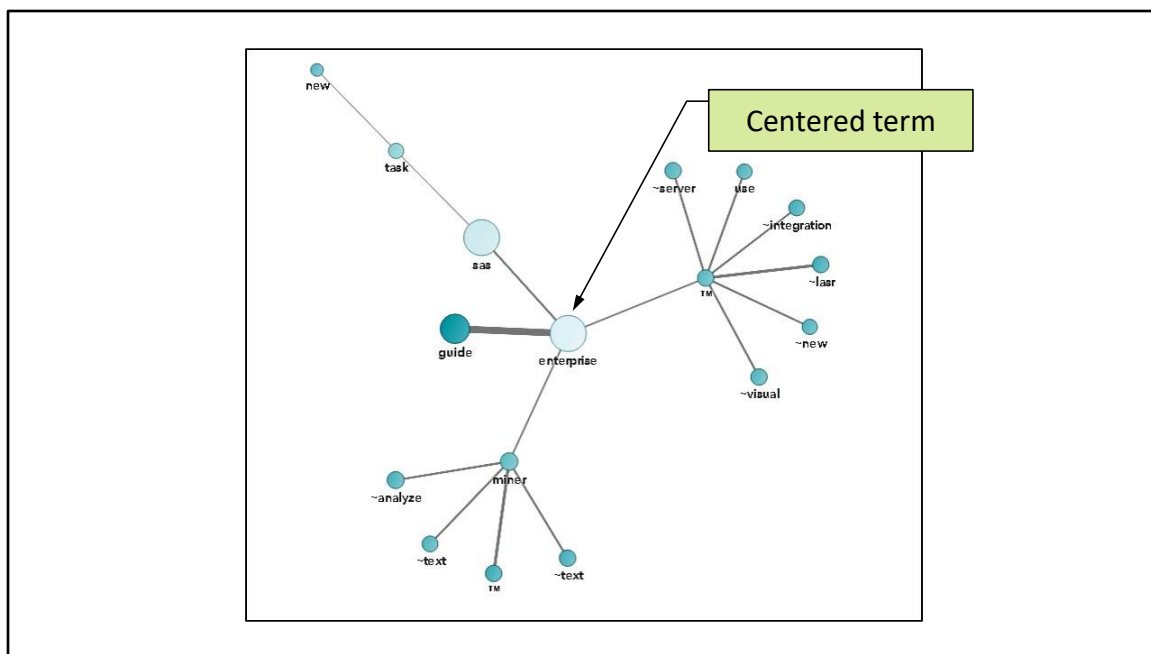
Words in the same document can be viewed as interacting directly. Words that do not appear together may be related because of shared direct relationships.



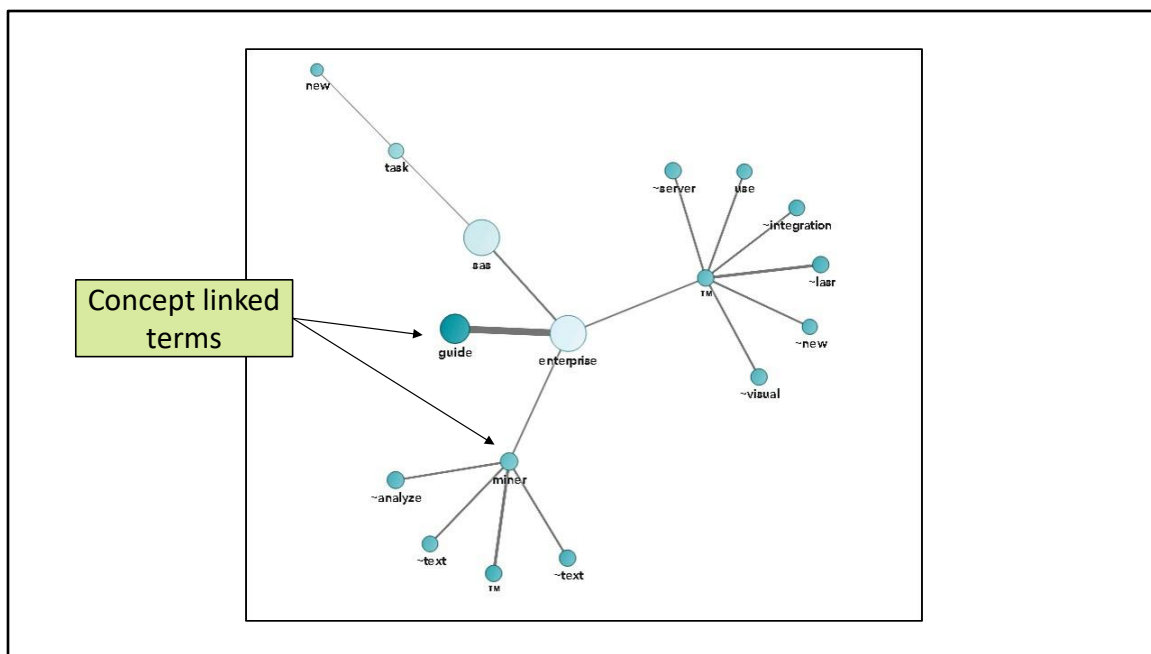
For example, one document may mention a car and a tire, while another mentions a truck and a tire, so car and truck are indirectly related through the word tire, and thus can be viewed as having an indirect social connection. This is called a *second order association*. The relationship depends on the context, and the word tire provides the context.



The Term Map window displays a term map for the selected term. The selected term,



called the centered term, is *enterprise*, and it is represented by the largest circle in the map. The term *guide* has the strongest association with *enterprise* based on the thickness of the line connecting the two terms.



Guide and *miner* are concept linked terms that co-occur with the centered term. For additional information about the term map display, click the **Help** icon in the term map display.

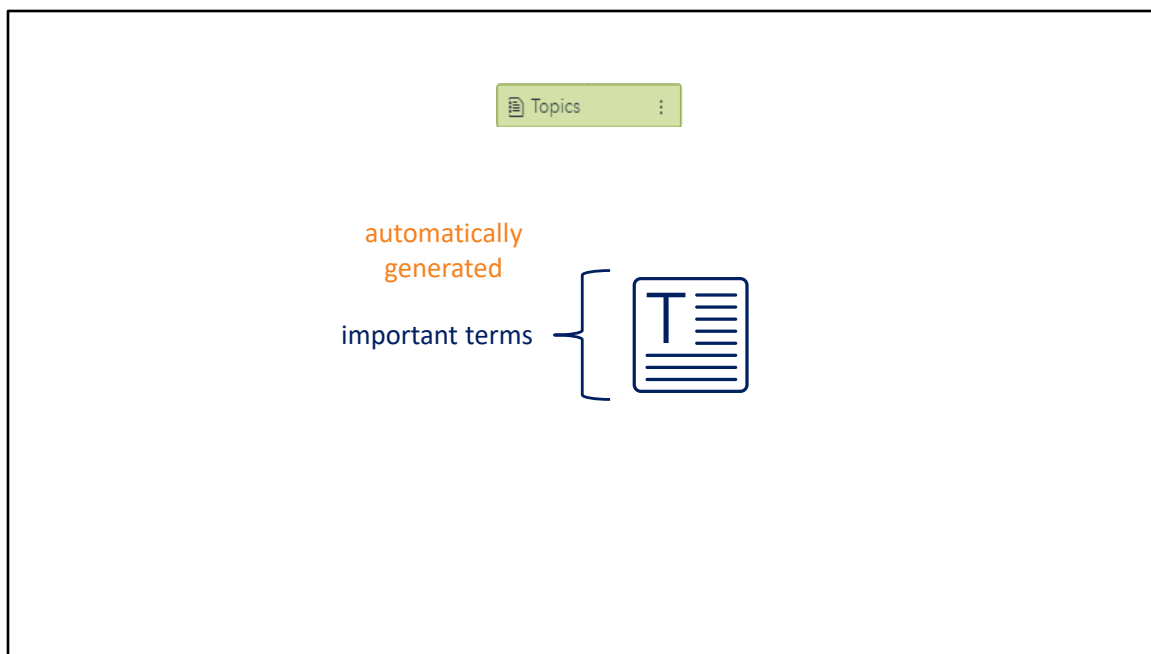
3.4 Machine-Generated Topics



Exploring Topics



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Topics are derived from natural groupings of important terms that occur in your documents. In SAS Visual Text Analytics, topics are automatically generated and assigned to documents. A single document can contain more than one topic.

Model Studio - Build Models

Drug Reports > Topics

Topics (10)

Topic	Created by	Documents
☒ +symptom, +withdrawal, off of, horrible, +dose	System	224
☐ +effect, +side effect, side, +side, sexual	System	223
☐ +work, ecstapin, best, +year, +try	System	223
☐ +day, +feel, mg, a day, +start	System	217
☐ abidal, +pain, just, +start, +work	System	208
☐ +medication, very, +good, +use, abidal	System	190
☐ +life, +save, +thought, +drugs, more	System	184

Terms

All (1522) Matched Filter

Term	Role	Documents	Frequency
☐ not	ADV	658	1174
☐ > take	V	677	1105
☐ > depression	N	492	616
☐ > feel	V	371	518
☐ > year	N	395	502
☐ > drug	N	341	485
☐ > day	N	344	479

Documents

All (1414) Matched (224 of 1414) Search

DrugReport	Sentiment	Relevancy
... taken ecstapin xr regularly for over 5 years, the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently went off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheaval. the side effects lasted...	⊖	0.334
... Ecstapin for about 5 years. The withdrawal symptoms are horrible. I have brain zaps , nausea , vomiting , dizzy , can't concentrate, I feel like I am in a another world and vision problems I slowly took myself off of this deadly drug . I went from 150mg to 75mg then to 37.5mg. I have been off it for about a week . My opinion with the withdrawals is like diazina your own grave. I wouldn't recommend this drugs to anybody.	⊖	0.330

Document 1 of 224

Highlight: Topic matches Search matches

The Topics page displays all the topics that SAS Visual Text Analytics identified. The default name of a topic is the top five terms based on relevancy scores. The terms table displays every term in the topic, its calculated relevancy, its assigned role, and the number of documents that contain that term. These terms are sorted in descending order based on their relevancy scores.

Document-level sentiments are also generated if they are preceded by a Sentiment node. The percentage of the documents in the topic that have a sentiment score of positive, negative, and neutral appears with each topic.

Model Studio - Build Models

Drug Reports > Topics

Topics (10)

Topic	Created by	Documents	Term	Role	Documents	Frequency
☒ +symptom, +withdrawal, off of, horrible, +dose	System	204				
☐ +effect, +side effect, side, +side, sexual	System	2		ADV	658	1174
☐ +work, ecstapin, best, +year, +try	System	2		V	677	1105
☐ +day, +feel, mg, a day, +start	System	217	☐ > feel	N	492	616
☐ abidal, +pain, just, +start, +work	System	208	☐ > year	V	371	518
☐ +medication, very, +good, +use, abidal	System	190	☐ > drug	N	395	502
☐ +life, +save, +thought, +drug, more	System	184	☐ > day	N	341	485
					344	479

Split a topic into two topics

Documents

All (1414) Matched (224 of 1414) Search

DrugReport	Sentiment	Relevancy
... taken ecstapin xr regularly for over 5 years, the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently want off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheaval. the side effects lasted...	⊖	0.334
... Ecstapin for about 5 years. The withdrawal symptoms are horrible. I have brain zaps , nausea , vomiting , dizzy , can't concentrate, i fill like i am in a another world and vision problems. I slowly took myself off of this deadly drug . I went from 150mg to 75mg then to 37.5mg. I have been off it for about a week . My opinion with the withdrawals is like diaoina your own arve. I wouldn't recommend this druo to anybody.	⊖	0.330

Document 1 of 224

Highlight: Topic matches Search matches

If a topic seems to convey information about two or more themes, you might want to split the topic to see whether the two themes emerge.

Model Studio - Build Models

Drug Reports > Topics

Topics (10)

Topic	Created by	Documents
☒ +symptom, +withdrawal, off of, horrible, +dose	System	224
☐ +effect, +side effect, side, +side, sexual	System	223
☐ +work, ecstapin, best, +year, +try	System	217
☐ +day, +feel, mg, a day, +start	System	208
☐ abidal, +pain, just, +start, +work	System	190
☐ +medication, very, +good, +use, abidal	System	184
☐ +life, +save, +thought, +drugs, more	System	184

Terms

All (1522) Matched Filter

Term	Role	Documents	Frequency
not	ADV	658	1174
	V	677	1105
	N	492	616
	V	371	518
> year	N	395	502
> drug	N	341	485
> day	N	344	479

Documents

All (1414) Matched (224 of 1414) Search

DrugReport

Text	Sentiment	Relevancy
... taken ecstapin xr regularly for over 5 years, the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently want off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheaval. the side effects lasted...	😊	0.334
... Ecstapin for about 5 years. The withdrawal symptoms are horrible. I have brain zaps , nausea , vomiting , dizzy , can't concentrate, i fill like i am in a another world and vision problems. I slowly took myself off of this deadly drug . I went from 150mg to 75mg then to 37.5mg. I have been off it for about a week . My opinion with the withdrawals is like diazina your own grave. I wouldn't recommend this drugs to anybody.	😊	0.330

Document 1 of 224

Highlight: Topic matches Search matches

Merge two or more topics

If you see two topics that seem related, you can merge them by selecting them and clicking **Merge Topic**. This action combines all the selected topics into the same topic.

The screenshot shows the SAS Model Studio - Build Models interface. The 'Topics' pane on the left lists 10 topics, each with a checkbox and a 'Documents' count. A green callout box with the text 'Add topics as categories.' points to a button in the 'Topics' pane. The 'Terms' pane on the right shows a table of terms, roles, documents, and frequency. The 'Documents' pane at the bottom shows a list of documents with sentiment and relevancy scores.

Topic	Created by	Documents
☒ +symptom, +withdrawal, off of, horrible, +dose	System	224
☐ +effect, +side effect, side, +side, sexual	System	223
☐ +work, ecstapin, best, +year, +try	System	223
☐ +day, +feel, mg, a day, +start	System	217
☐ abidal, +pain, just, +start, +work	System	208
☐ +medication, very, +good, +use, abidal	System	190
☐ +life, +save, +thought, +drugs, more	System	184

Term	Role	Documents	Frequency
not	ADV	658	1174
		677	1105
		492	616
		371	518
> year	N	395	502
> drug	N	341	485
> day	N	344	479

Document	Sentiment	Relevancy
... taken ecstapin xr regularly for over 5 years, the side effects of going off of it (missing from one dose to several) are so severe i get anxiety just thinking about it. i recently went off of it suddenly and the withdrawal symptoms ranged from the most severe nausea i've ever experienced (including vomiting) to hyper-sensitivity through my fingers to a brain-clicking feeling like a cd that keeps skipping to anxiety and emotional upheaval. the side effects lasted...	😊	0.334
... Ecstapin for about 5 years. The withdrawal symptoms are horrible. I have brain zaps, nausea, vomiting, dizzy, can't concentrate, I feel like I am in a another world and vision problems I slowly took myself off of this deadly drug. I went from 150mg to 75mg then to 37.5mg. I have been off it for about a week. My opinion with the withdrawals is like diazina your own grave. I wouldn't recommend this drug to anybody.	😞	0.330

The next step in the analysis process is to identify which topics you want to promote to categories. To promote a topic to a category, select that topic in the Topics pane and click **Add topics as categories**. SAS Visual Text Analytics adds the selected topic to the Categories page. You can promote multiple topics to categories at one time.

>> Topics

Description:

Assigns documents to topics.

▼ Topic Discovery

☒ Automatically determine number of topics

Maximum topics:

25

Term density:

0 1 5 10

Document density:

0 1 5 10

You can edit the properties that affect all topics by changing the settings for the Topics node. Term density refers to how topics are populated with terms. It's defined by a number between 1 and 10. The default is 1. Increasing the term density number captures fewer documents.

This value affects the number of documents that belong to a topic. For example, having fewer terms in a topic captures fewer documents. Values that you enter are rounded to the nearest integer.

You can also designate a maximum number of topics that you want generated for the project. If you do not enter a value, the software uses default methods to generate topics from important terms.

Run the topics to see the results of your changes.

Topics (16)			Terms				
☐ Topic	Created by	Documents ↓	All (21918) Matched (13 of 21918) Filter				
☒ taxiway, runway, taxi, clearance, groundcontrol	User	16117	☐ Term	Relevancy ↓	Role	Documents	Frequency
☐ taxiway, runway, taxi, clearance, tower	User	4176	☐ crossrunway	1.000	_JNCURSION	546	844
☐ tower, takeoff, clearance, runway, hear	System	3335	☐ taxiway	1.000	_JNCURSION	2339	8705
☐ flap, knot, brake, +autopilot, gear	System	3222	☐ cross	1.000	_JNCURSION	2704	5113
☐ approach, runway, visual, visual approach, +locate	System	3004	☐ ramp	1.000	N	1012	1748
☐ +foot, descend, altitude, +autopilot, cross	System	2989	☐ taxi	1.000	_JNCURSION	3662	7700
☐ engine, emergency, declare, smoke, checklist	System	2869	☐ groundcontrol	1.000	_JNCURSION	1135	1982
☐ minimumequipmentlist, takeoff, flap, dispatch, flight	System	2867	☐ hear	1.000	V	2988	4509
			☐ control	1.000	N	7020	15306

You can select a subset of terms in a derived topic to create a custom topic. The custom topic assigns a term weight of 1 to each term. You can merge two topics to form a new topic. Then you can select terms from the merged topic to create a custom topic. This provides flexibility when you are creating custom topics. You can use subject-matter expertise to build customized dictionaries. Although a custom topic acts much like a custom concept that uses CLASSIFIER rules, the results can be different. This is because concepts apply Boolean operations, where topics apply numeric scoring.

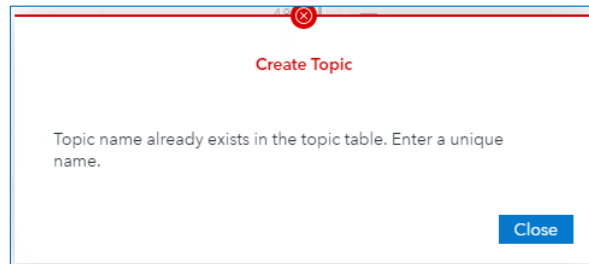
Select terms

Create a topic from terms

Term	Role	Documents	Frequency
☒ depression	N	492	616
☒ depressive	A	17	18
☒ depressive	PN	4	4
☒ depressive	N	4	4
☒ depressive episode	nlpNounGroup	4	5
☒ depressive symptom	nlpNounGroup	4	4
☐ describe	V	16	16
☐ desire	N	12	12

Term	Relevancy ↓	Role	Documents	Frequency ↓
☐ depressive	1.000	N	4	4
☐ depression	1.000	N	492	616
☐ depressive	1.000	A	17	18
☐ depressive symptom	1.000	nlpNounGroup	4	4
☐ depressive episode	1.000	nlpNounGroup	4	5
☐ depressive	1.000	PN	4	4

To create a custom topic, select all the terms that define the topic by clicking the square to the left of each term in the Terms pane. After you have selected all the terms that apply, click the **Create a topic from terms** icon in the upper right of the Terms pane. A new topic will be added to the Topics pane having the name of five of the selected terms. All terms in the custom topic will have a relevancy score of one.



Unfortunately, if you select the first five terms that define an existing topic, you will not be allowed create a custom topic. The following error message appears: Topic name already exists in the topic table. Enter a unique name.

To overcome this unfortunate limitation, you can try to emulate the desired topic by creating separate topics, and then joining them. A new set of relevancy values will be derived for the joined topic, so you will not get relevancy weights equal to one. Since you cannot delete topics, this approach often leads to a surplus of topics that will increase the run time of a pipeline.

Term and Document Density

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



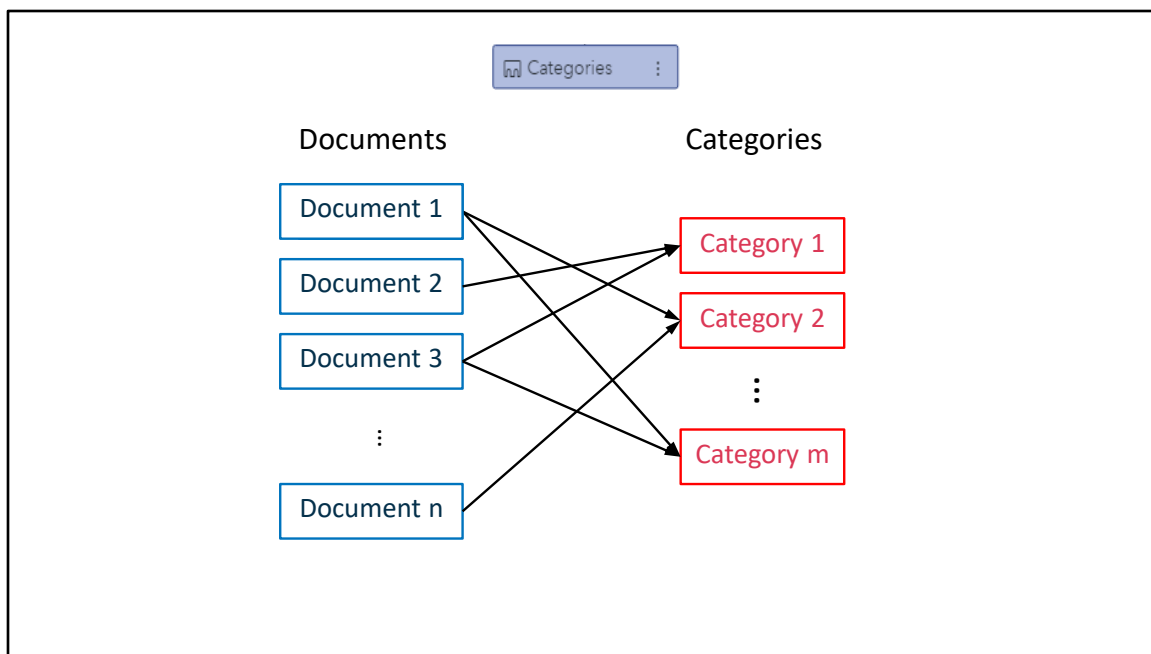
3.5 Categories



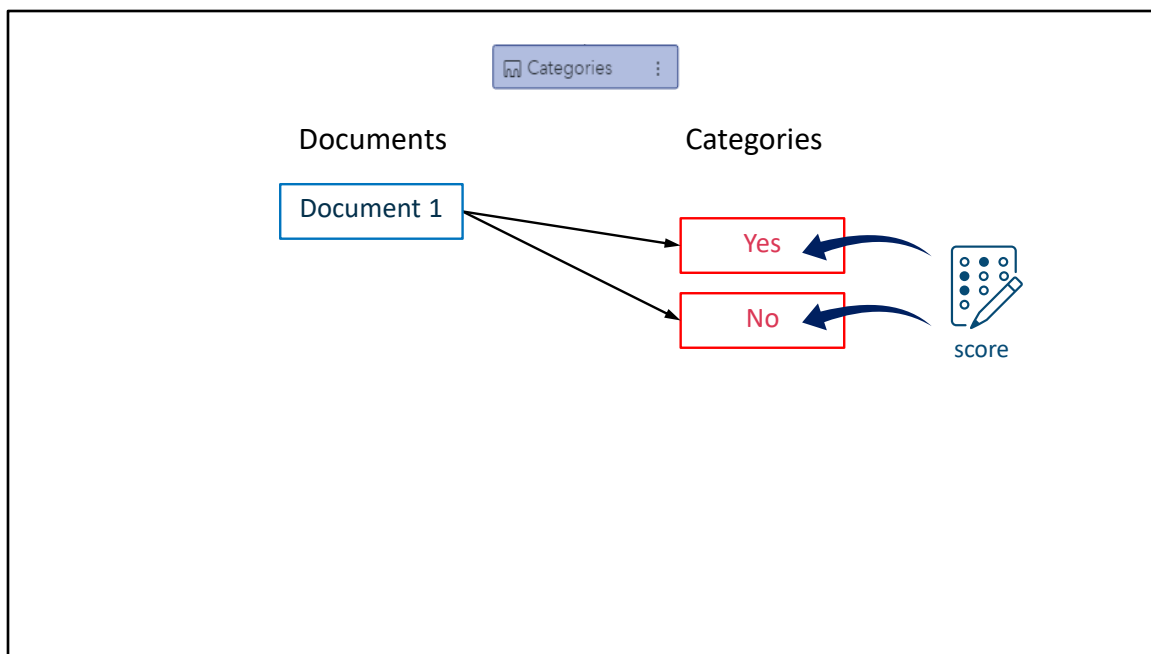
Reviewing Category Rules and Results



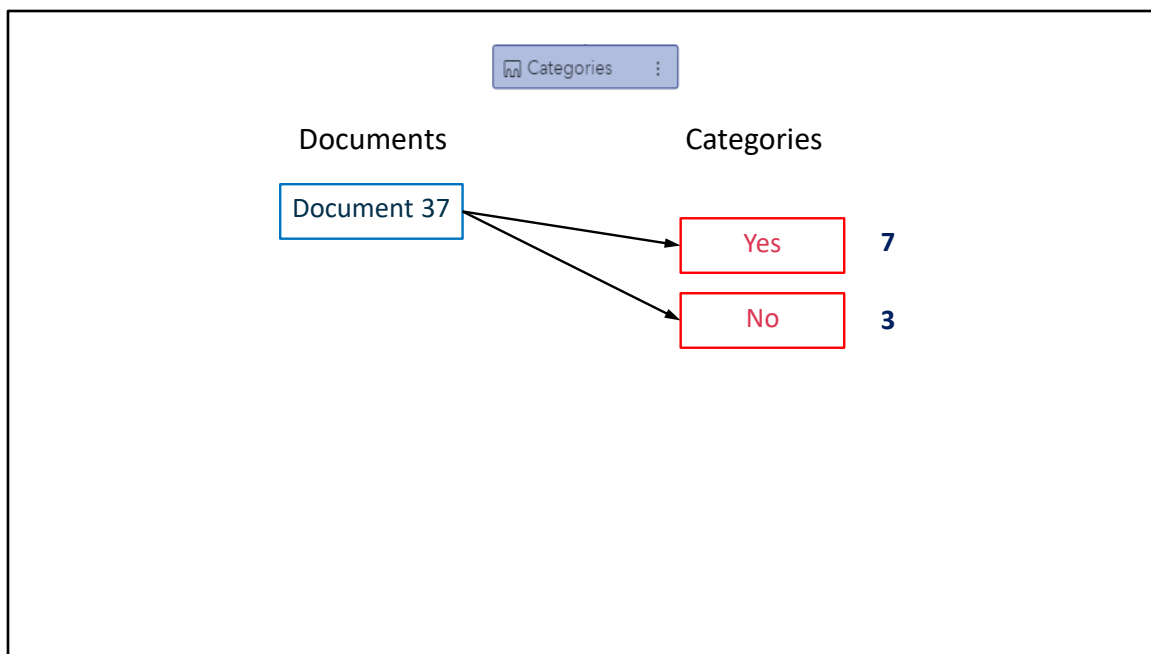
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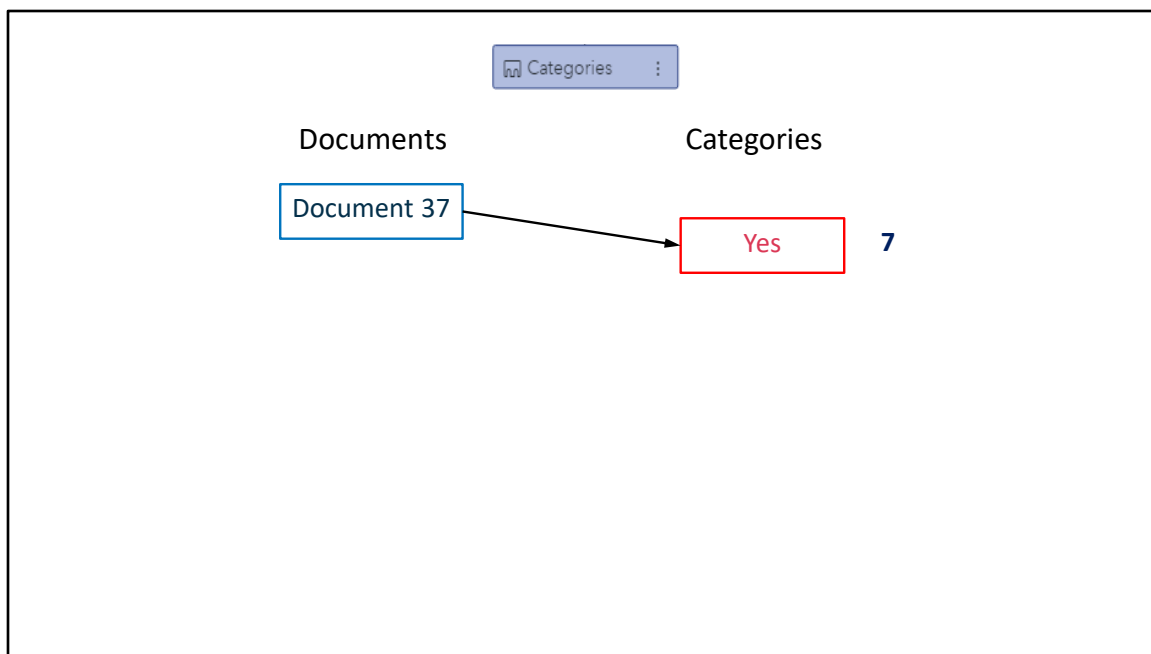
The Categories node allows a document to fall into more than one category.



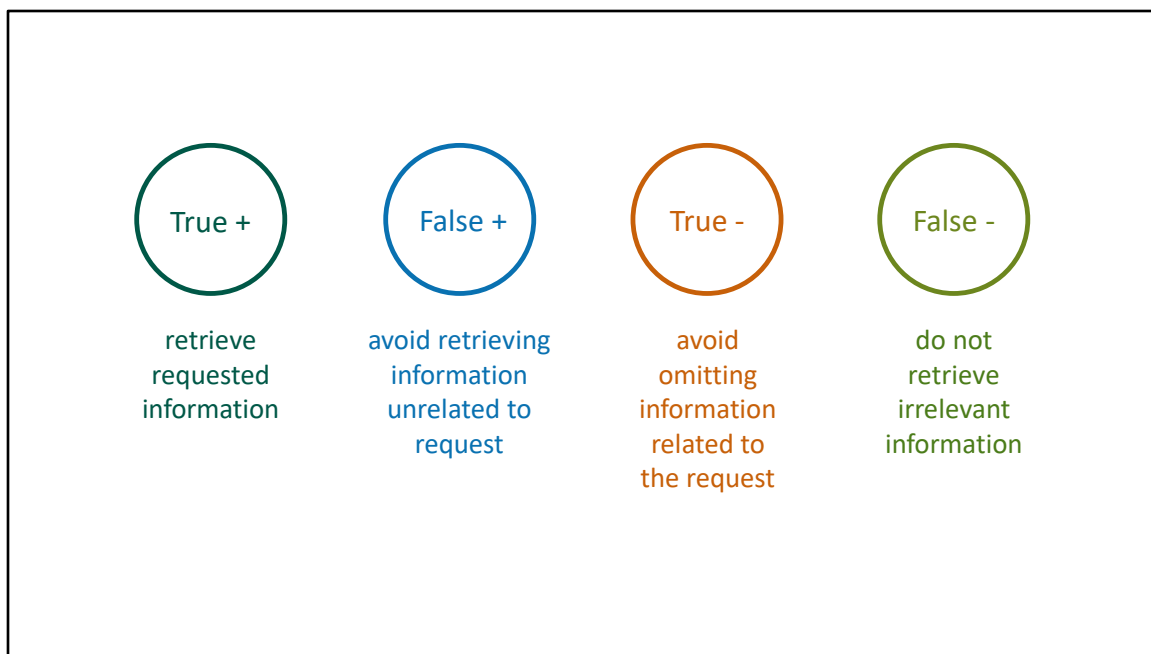
If a document is classified as belonging to both a Yes and No category for a binary category variable, a score is assigned to each category so that ambiguities can be resolved.



Suppose document 37 is assigned to the Yes category with a score of 7 and to the No category with a score of 3. Since the Yes category score is higher,



you can choose to assign document 37 to the Yes category.



When a category rule is based on a category target variable, you can determine the accuracy of the classification. There are four possible outcomes from a classified document: true positive, false positive, true negative, and false negative. For a true positive, you want to retrieve the information that was requested. For a false positive, you want to avoid retrieving information that is unrelated to the request. For a true negative, you want to avoid omitting information that is related to the request. And for false negatives, you do not want to retrieve information that is irrelevant.

Notation

ND = Number of documents

NS = Number of selected documents (retrieved)

TP = Number of true positives

FP = Number of false positives

TN = Number of true negatives

FN = Number of false negatives

		Action	
		Retrieved	Omitted
Actual	Contain Info	TP	FN
	Absent Info	FP	TN
Column Totals		NS	ND-NS

Category rules can be assessed based on summary statistics derived using the four possible outcomes.

Misclassification Rate = Fraction
misclassified

$$\text{_MISC_} = (FP + FN)/ND$$

Precision = Fraction of retrieved
documents that are relevant

$$\text{Precision} = TP/(TP + FP)$$

Recall = Fraction of relevant
documents that are retrieved

$$\text{Recall} = TP/(TP + FN)$$

		Action	
		Retrieved	Omitted
Actual	Contain Info	TP	FN
	Absent Info	FP	TN
Column Totals		NS	ND-NS

The misclassification rate, often presented as a percentage, is the fraction of documents that are incorrectly classified. Precision is the fraction of documents that are classified as positives that are true positives. Recall is the fraction of all positive documents that are classified as positive.

F1 Statistic = Harmonic mean of precision and recall

$$F1 = 2 / [(1/Precision) + (1/Recall)] = (2 * Precision * Recall) / (Precision + Recall)$$

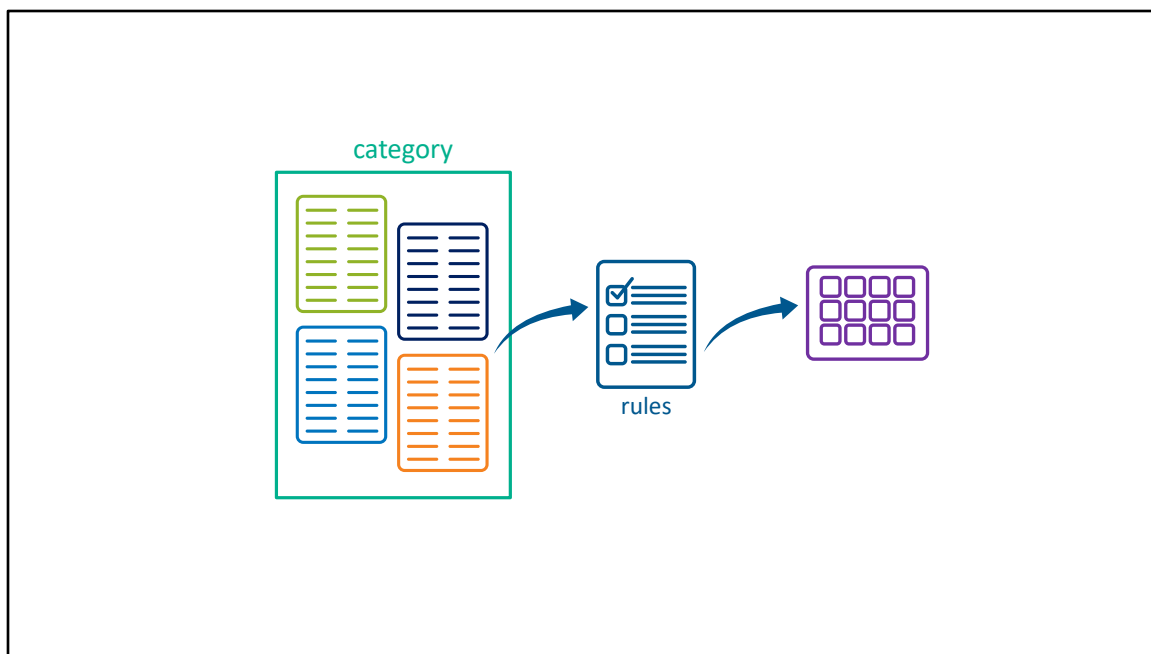
(F1 is also called the *F-Measure*.)

The F measure, or F1 statistic, is the harmonic mean of precision and recall and is one way to measure the tradeoff between accurate information retrieval and comprehensive information retrieval.

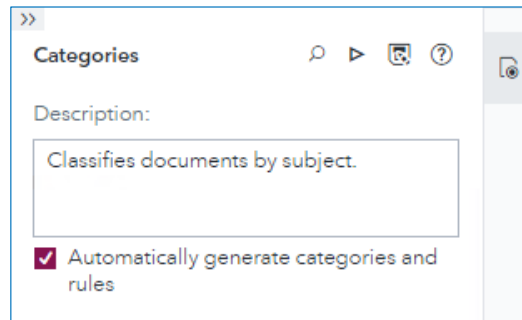
Exploring the Category Node



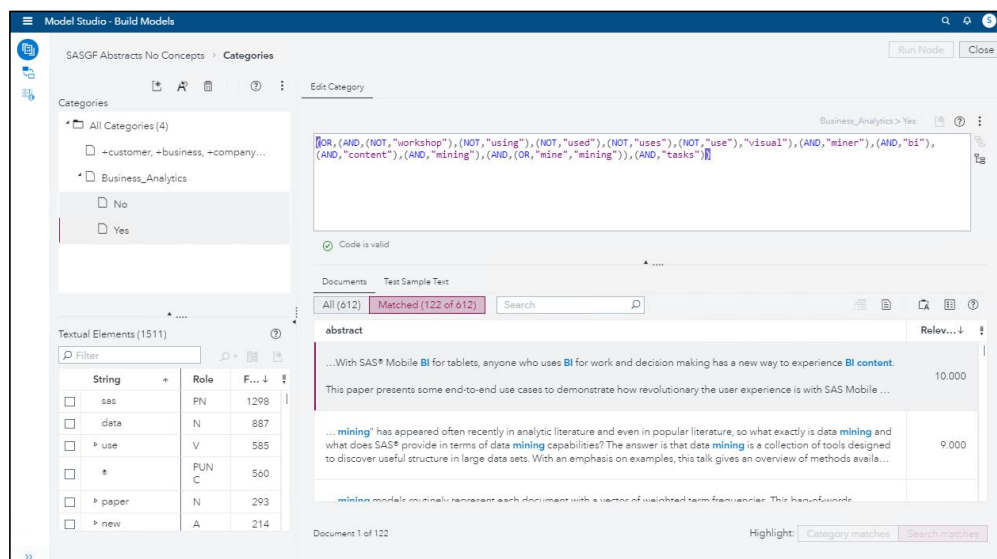
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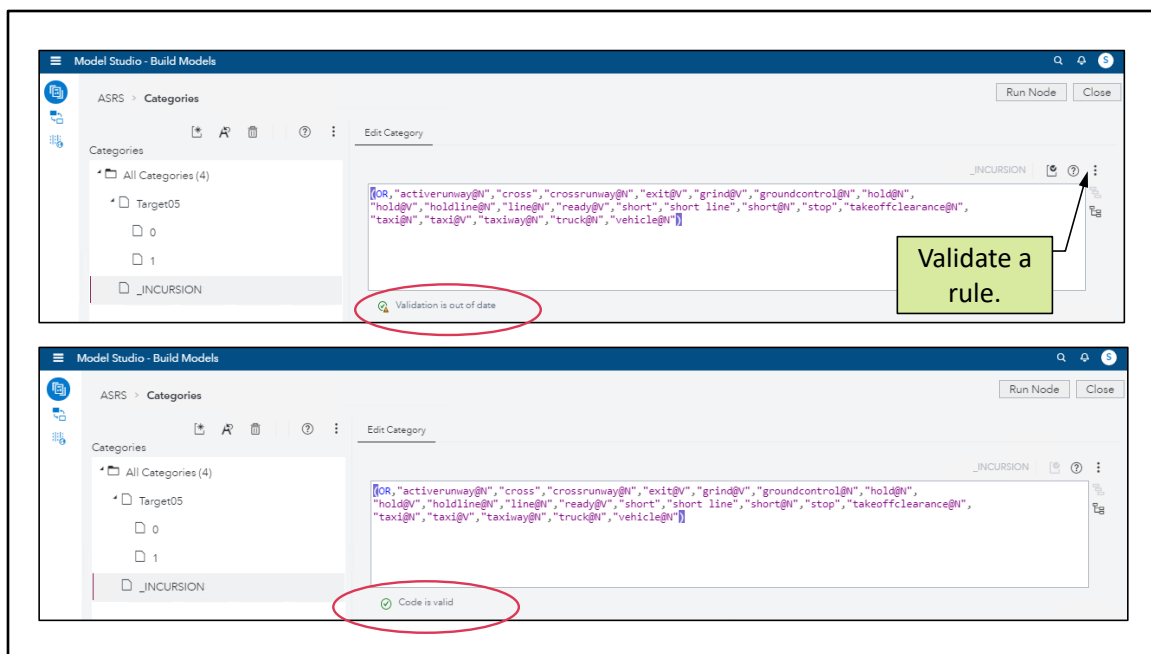
A *category* identifies a group of documents that share a common characteristic. The rules for each category are executed against the input data set.



The Categories node has a single property that determines if linguistic rules will automatically be derived for all category variables.



The Categories node uses machine learning and natural language processing to derive category rules for each category variable. The rules can be understood without any prerequisite knowledge of NLP algorithms. The rules can also be edited to incorporate domain knowledge that is not available to the software.



You can validate manually edited or created rules to ensure that there are no syntax errors.

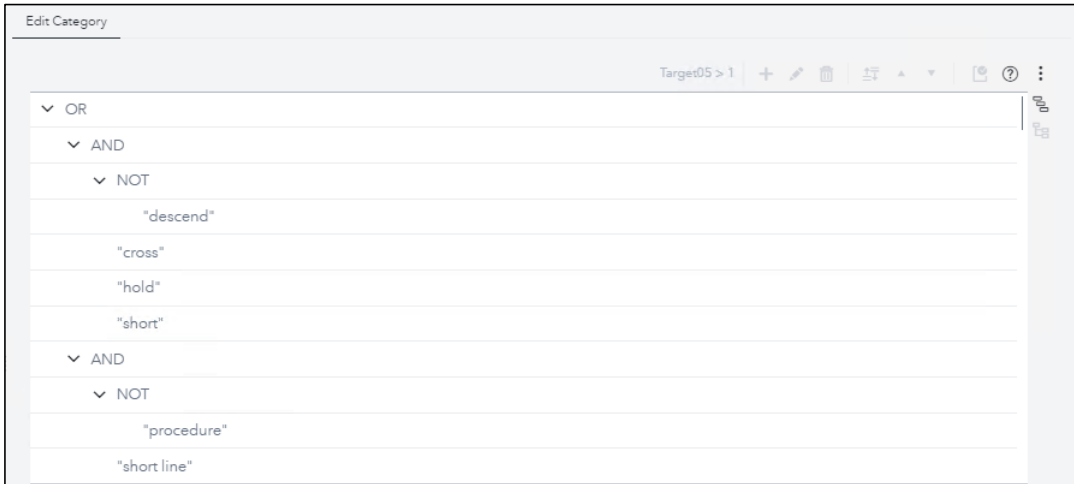
The screenshot shows a SAS interface for testing a rule. At the top, there are tabs for 'Documents' and 'Test Sample Text'. Below the tabs, there are filters for 'All (21519)' and 'Matched (11658 of 21519)', along with a search bar. A yellow box labeled 'Test a rule.' points to the 'Test Sample Text' tab. The main area displays a table with three columns: 'Text', 'Sentiment', and 'Relevancy ↓'. The table contains three rows of text, each with a sentiment icon (a sad face) and a relevancy score. A pop-up window shows the rule being tested: 'Groundcontrol cleared me to cross runway two right, but I had to stop for a fuel truck.'

Text	Sentiment	Relevancy ↓
taxi JUST bare PAST THE instrumentlandingsystem HOLD SHORT LINE AT approach END OF runway AT sdf airport._ aircraft WERE SENT AROUND.THIS occur FOR sever REASON _ THE control say TAXI TO runway HOLD SHORT OF THE approach.NOT HOLD SHORT OF...	☹️	130.000
cross OF HOLD SHORT LINE FOR runway _WE WERE clear TO THE hold PAD FOR runway _ AT dca airport TO WAIT OUT A ground STOP FOR ORD.AFTER approximate _ hour WE WERE give A reroute AND TOLD TO START engine AND TAXI TO HOLD SHORT OF runway _I THOUGHT...	☹️	107.000
...WAS clear TO TAXI bywayof taxiway P taxiway left FOR A runway depart.AS I near taxiway south ON taxiway left I WAS ask IF I WAS FAMILIAR WITH ORANGE transit I state I WAS UNFAMILIAR AND WOULD LIKE A progressive	☹️	104.000

Document 1 of 11658

Groundcontrol cleared me to cross runway two right, but I had to stop for a fuel truck.

You can test a rule to ensure it's capturing the information as intended. The highlighted terms show how the rule is invoked for a specific test case.



The tree view makes it easier to decipher the logic behind a rule. The hierarchy of the Boolean rules is made clear by the tree formatting. For the rule given, a NOT rule is within each AND rule, and the two AND rules are within the parent OR rule.

3.6 Document Scoring



Scoring New Documents

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Resource

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